Name _________________

Homework Activity for Molecules and Life Week 2
Due the week of November 10
SHOW ALL WORK!

1. <u>Gene Editing: CRISPR-Cas9</u>

In lecture, we learned about CRISPR-Cas9, a novel technique for editing genes.

1A. (i) How was CRISPR-Cas9 discovered? Were researchers aiming to find a new technique for editing genes? 2-3 sentences are sufficient.

CRISPR is also used as a powerful tool in clinical research. Let's explore an example from cancer research.

BRCA1 and BRCA2 normally help repair damaged DNA, but mutations in these genes impair DNA repair and increase cancer risk. Chemotherapy, the standard form of cancer therapy, increases DNA damage in cancer cells, thereby preventing replication and triggering cell death. The drug Olaparib targets another DNA repair protein called PARP and is effective only in cancers with BRCA mutations, since blocking both repair pathways causes cancer cells to die.

To investigate the relationship between Olaparib and BRCA mutations, researchers used CRISPR-Cas9 to introduce BRCA mutations into pancreatic cells in the lab. These mutations introduced premature STOP codons to make the cells produce less functional BRCA protein. By doing this, scientists were able to compare how well Olaparib worked in cells with and without BRCA mutations.

1A. (ii) Which of the following accurately describes Cas9 and its function in genome editing?

a. A protein responsible for DNA methylation.
b. A virus that infects bacteria and introduces new genetic material.
c. A protein enzyme that cuts both strands of DNA at sites specified by an RNA guide.
d. A small RNA molecule that serves as a template for DNA replication.

1A. (iii) What effect does incorporating a premature STOP codon into the BRCA gene have on the BRCA proteins that are expressed from that gene?

a. Incorporating the STOP codon deletes the entire BRCA gene, eliminating all BRCA protein production.
b. Incorporating the STOP codon into the BRCA gene sequence stops translation early and leads to the production of a truncated version of BRCA protein that is non-functional.
c. Incorporating the STOP codon enhances the production of fully functional BRCA proteins.
d. Incorporating the STOP codon has no impact on the translation process, as the cellular machinery ignores the premature STOP signal

While CRISPR-Cas9 is a powerful gene-editing tool, it is not without limitations. In the following question, we will examine the likelihood of off-target effects as well as editing success.

1B. (i) In order to guide the CRISPR-Cas9 complex to the BRCA genes, the researchers designed a 20-base pair (bp) guide RNA. **How many possible unique 20-bp guide RNA sequences exist?**

1B. (ii) A 20-bp sequence is used to guide CRISPR-Cas9 to its target site in the pancreatic cells that the researchers are aiming to edit. **What is the probability that a randomly selected site is an exact match for a given 20-bp gRNA sequence?** Assume all possible sequences are equally likely.

1B. (iii) In order to introduce premature STOP codons into the BRCA1 and BRCA2 genes, the researchers need to edit 4 sites in the DNA. In previous trials, the researchers found that their genome editing procedure could successfully edit a single site in a single cell 85% of the time. **Based on this information, estimate the probability that they can successfully edit all four target mutations in one pancreatic cell.**

1B. (iv) Estimate the probability that they can successfully edit all four target mutations in every cell in a pool of 50 pancreatic cells.

1B. (v) What is the probability that they can successfully edit at least one of the four target mutations in every cell in a pool of 50 pancreatic cells? [Hint: $P = 1-(1-P_{one})^n$]

After performing the CRISPR treatment, the researchers aimed to determine whether the number of successful edits was sufficient to reduce the level of BRCA proteins. To do this, they measured BRCA protein levels in three groups of cells: untreated wild-type (WT), CRISPR-Cas9 control (CC_ctrl; treated with Cas9 but without a donor sequence), and CRISPR-Cas9 treated cells (CC_BRCA1 or 2) that received the donor sequence containing the stop codon.

1C. (i) Why did the researchers include the CRISPR Cas9 control (CC_ctrl) in the experiment?

1C. (ii) Based on the graph below, evaluate whether gene editing successfully reduced BRCA protein levels. Use your knowledge of statistics and make sure you analyze all the given comparisons (six in total).

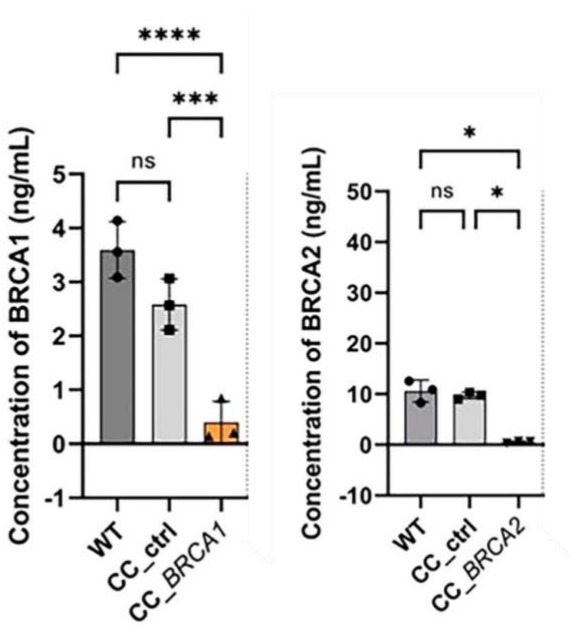

Figure 1. Analysis of BRCA1 and BRCA2 protein concentration. Data are represented as the mean values and standard deviation (SD) of 3 independent experiments for non transfected (wild-type (WT)) cells and CRISPR/Cas9 control (CC_ctrl) (a CRISPR/Cas9 complex without donor sequences) and CC_BRCA1 or BRCA2 - transfected cells.
*****P<0.0001, ***P<0.001, **P<0.01, *P<0.05 and ns not significant.*

With these pancreatic cell lines in hand, the researchers set out to test whether cells with BRCA mutations show increased sensitivity to, or are more strongly affected by, the drug olaparib.

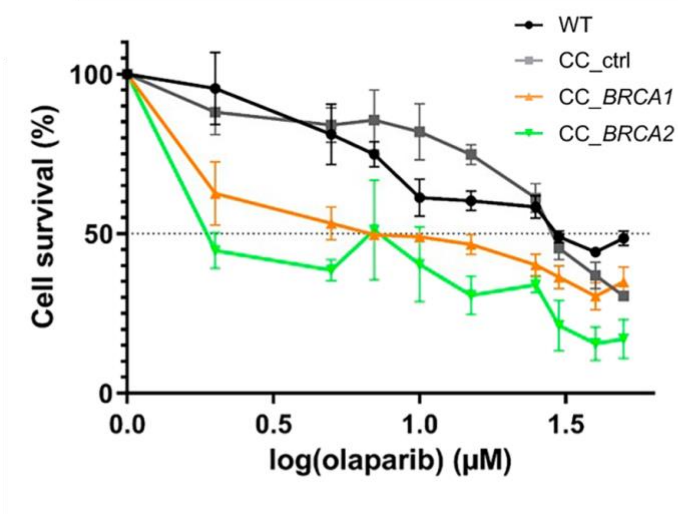

Figure 2. *Percentage of cell viability versus logarithm of the concentration of olaparib obtained for untreated cells (wild-type (WT), and cells treated with a CRISPR/Cas9 control (CC_ctrl, a CRISPR/Cas complex without donor sequences), a CC_BRCA1 and a CC_BRCA2. Increasing concentrations of olaparib (0–50 μM) were applied for 72 h. Error bars represent mean ± 2SE.*

1C. (iii) Analyzing the data presented in Figure 2, determine whether pancreatic cells with BRCA mutations show increased sensitivity to olaparib. Support your answer with your knowledge of statistics.

1C. (iv) What effect does increasing the concentration of olaparib have on the survival rate of wild-type (untreated) cells? Why might this be the case?

1C. (v) Given what you know about how Chemotherapy/Radiation therapy works, why do patients with BRCA mutations tend to respond better to DNA-damaging therapies like olaparib?

2. <u>Molecular Basis of Cancer</u>

In this week's lecture you also learned about the evolution of cancer and specific genes that are involved.

2A. (i) Which of the following statements best describes the genetic evolution of cancer?

a. Cancer is caused by a single mutation in a cell that leads to uncontrolled growth.
b. Cancer develops through a series of mutations that accumulate over time, affecting genes involved in cell growth, division, and repair.
c. Cancer cells are genetically identical to normal cells and differ only in their behavior.
d. Cancer is primarily caused by environmental factors and is not much influenced by genetics.

2A. (ii) You are working in a cancer biology lab. You've just been handed four gene profiles based on the tumor samples of various patients. Each summary describes what the gene does or how it behaves when mutated in cancer. Your task is to classify each gene into one of the following categories: **Tumor Suppressor/ Proto-Oncogene / DNA Repair Gene.**

Gene A:
This gene encodes a protein that is activated in response to DNA damage. When functional, it halts the cell cycle and may trigger cell death. In many tumors, it is deleted or mutated.

Gene B:
This gene encodes a protein that helps control translation of other proteins. When too much of the protein is made, it results in overproduction of proteins that promote cell growth, which may lead to cancer.

Gene C:
This gene encodes a protein that checks newly replicated DNA for mismatches and initiates repair. Mutations lead to additional errors in DNA, which can lead to certain types of cancer.

Gene D:
This gene helps control how much and how often cells grow by turning off certain signals. When the gene is mutated, those signals stay on, which can cause cells to grow too quickly or uncontrollably.

Your supervisor in the lab has selected Gene B to study further. You want to assess the safety and efficacy of a potential chemotherapy drug that targets the protein product of Gene B using laboratory mice. When each mouse is 36 days old, it is given either an injection containing water, or an injection containing cells with cancer-causing mutations to Gene B that can implant into the host mouse and grow into tumors.

All mice are then grouped into one of five drug treatments, each corresponding to a given medication dose, as listed in Table 1, and are injected with that dose once per week for ten weeks. The lifespan of each individual mouse is recorded, producing the data in Table 1 below:

Table 1. *Mean mouse lifespan (in days) for water- and cancer- injected mice treated with one of five dosages of experimental chemotherapy medication.*

Medication Dose (ng)	Water-injected mice		Cancer-injected mice	
	Mean	**95% CI**	**Mean**	**95% CI**
0	660	649-671	268	248.5 - 287
75	618	610 - 625	289	276 - 301.5
150	515	500 - 529	353	342 - 364
300	495	486.5 - 503	387	370 - 404
600	405	392 - 417.5	392	378.5 - 405

2B. (i) What is one control treatment included in this experimental design and what is it controlling for? Select all that apply.

a. 0 ng of medication as a control stimulus to establish a baseline lifespan for comparing cancerous and non-cancerous mice.
b. Non-cancerous mice injected with water to control for drug safety and its physiological effects separate from cancer.
c. High dosage group to determine maximum drug efficacy.
d. Treating multiple mice in each group to account for individual differences among mice.

2B. (ii) Based on your answer to question 2A. (ii) what effect might this medication be having on the protein expressed from Gene B? Select all that apply**.**

a. The medication would inhibit the activity of the protein.
b. The medication would increase the activity of the protein.
c. The medication would decrease the amount of protein expressed.
d. The medication would increase the amount of protein expressed.

2B. (iii) Based on these findings, which dosage(s) is/are most effective at fighting cancer? Which are the least damaging to healthy mice? Explain using your knowledge of statistics.

2B. (iii) Chemotherapy is damaging to both cancer cells and healthy cells. The most effective dosage requires balancing the potential benefits of the treatment against its potential costs. **Which dosage would you choose to move forward with for tests in human cancer patients?** Explain your answer in 2-3 sentences.

3. <u>Midterm Exam Wrapper</u>

A. How did you prepare for the midterm exam (for example, reviewing concept learning objectives, reviewing homework assignments, re-doing Argos lessons, reading the Tutorials etc.)?

B. Approximately how much time did you spend preparing for the midterm exam?

C. Which aspects of the course were most helpful in your midterm preparation (attending lecture, attending seminar, completing Argos, completing homework assignments, etc.)?

D. Do you feel that the concepts and habits covered in the midterm were adequately addressed in the seminar? If not, which ones?

E. What type of questions was easiest for you to answer: questions related to content covered in the lectures; calculation-based questions or questions involving habits (reading graphs, log plots, experimental design, statistics, etc.)? What type of questions was more difficult for you?

F. Did you have enough time to answer all of the questions and to check your work?

G. How do you think that you should study for the final exam in order to maintain your level of knowledge and to improve in the areas that were difficult for you?

H. What can I do as your seminar leader to help you better prepare for the final exam?